

Shaik Mahammad Khaleel

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SUMMARY

- Hands on experience in synthesis, STA, Floor Planning, Power Planning, Placement, CTS, Routing and IR Drop Analysis.
- Have solid experience in Logical and Physical Synthesis, Macro Placement, Routing at advanced Node, Static Timing Analysis, Timing Closure techniques, Noise analysis, Functional Eco's, Metal Eco's implementation.
- Experience in lower technology nodes such as TSMC 16nm, 28nm and 32nm able to handle complex designs with multimillion gates and tens of macros.
- Having better understanding of clock tree structure and implementation.
- Hands on experience in low power implementation techniques such as ON-OFF and Multi-Voltage designs.
- Have hands on experience in DRC, LVS, LEC, DRV fixes, Crosstalk analysis, Glitches, Antenna and EM/IR fixes.
- Strong interest and understanding of Advanced Node and Design methodologies.
- Proficient in scripting languages TCL, PERL and good at UNIX.

WORK EXPERIENCE

- One year internship as a Physical Design Engineer at RiseTime semiconductors (OPC) Pvt Ltd, from 4th April, 2022 to 22nd April, 2023.

ACADEMIC QUALIFICATIONS

Qualification	Specialization	University/Board	Institute	Year	CGPA/Percentage
Post Graduate Certification Program	PHYSICAL DESIGN-VLSI	JAIN	JAINX	2022-2023	B+
B.Tech	EEE	JNTUK	Tirumala Engineering college	2020	8.34 CGPA
Diploma	EEE	AP SBTET	A.M. Reddy Memorial college	2017	85.27%
Matriculation	SSC	Board of Secondary Education Andhra Pradesh	B R C high school	2014	9.2 CGPA

TOOLS

Synthesis	Design Compiler, Genus and Fusion Compiler
Place and Route	ICC2 and Fusion Compiler
Timing Analysis	Prime Time
Logical Equivalent Check	Formality

PROJECT DEATILS

BLOCK I

Instance Count	: 2.2M	Number of Macros	: 16
Technology node	: TSMC 28nm	Metal Layers	: 9
Number of Clocks	: 2	Operating Frequency	: 1.2 GHz,
520MHz			
Tools used	: ICC2, PT, Formality.		
Challenges faced	: Congestion issues, local density, clock latencies & skew.		
Role Played	: Floorplanning, Place and Route, DRC clean, STA Signoff.		

BLOCK II

Instance Count	: 2.8M	Number of Macros	: 186
Technology node	: TSMC 16nmX9M	Operating Frequency	: 602MHz
Number of Clocks	: 4		
Challenges faced	: IR fixes		
Tools used	: ICC2, Prime Time, Formality.		
Role Played	: Performed Signoff Timing analysis, Calibre DRC & LVS and Formality checks.		

BLOCK III

Instance Count	: 6.2k	Number of Macros	: 4
Technology node	: TSMC 16nm	Metal Layers	: 9
Number of Clocks	: 1	Operating Frequency	: 150MHz
Challenges faced	: Congestion, fixing max cap, max trans, IR drop analysis DRC and LVS.		
Tools used	: ICC2, Prime Time, Formality.		
Role Played	: Sanity checks, Floorplanning, Placement, CTS, Routing, Timing analysis and Signoff.		

DECLARATION

I do hereby declare that the above particulars of facts and information stated are true, correct, and complete to the best of my belief and knowledge.

Place : Hyderabad

Signature

Date : 03 May, 2023

Shaik Mahammad Khaleel